



CONDUCTIVITY — FULL MIND MAP

चालकता · Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this chapter:

1. Conductors, Free Electrons & Electrical Conductivity
2. Thomson Effect, Faraday Cage & Electronic Diodes
3. Superconductivity & High-Temperature Superconductors
4. Semiconductor Behaviour, Charge Carriers & Transistor Materials

1. Conductors, Free Electrons & Electrical Conductivity

- Conductor/सुचालक me large number of free electrons hote hain, isliye electrical resistance low aur current flow easy hota hai. Insulator me free charge carriers bahut kam hote hain. [Q1]
- Given common metals me silver sabse achchha electrical conductor hai; source ranking repeatedly Silver > Copper > Aluminium > Iron ko support karti hai. Mica insulator hai. [Q2–Q4]
- Salt water, orange juice aur lemon juice ions/acids-salts ki wajah se electricity conduct karte hain; isliye source Q5 me “None of the above” bad conductor answer hai. [Q5]
- Lightning conductor ke liye copper iron se preferred hota hai because copper ki electrical conductivity higher hai aur atmospheric corrosion/rusting comparatively kam hota hai. [Q6]

⚠ EXAM TRAP — Q6 — Lightning Conductor Reasoning

- ▶ Primary exam point: copper high electrical conductivity + good corrosion resistance ki wajah se preferred hai.
- ▶ Source explanation me high thermal conductivity ko “energy loss less” se joda gaya hai; lightning protection ka core electrical reason low-resistance conducting path hai.

2. Thomson Effect, Faraday Cage & Electronic Diodes

- Ek hi conductor ke different points par temperature gradient ho aur current flow kare, to reversible heating/cooling associated thermoelectric phenomenon ko Thomson effect kehte hain. [Q7]
- Closed metallic car lightning ke time Faraday cage jaisi protection de sakti hai: excess charge outer conducting surface par redistribute hota hai aur cabin ko strong electric field se shield karta hai. [Q8]
- Tunnel diode heavily doped semiconductor junction hota hai aur quantum tunnelling ki wajah se extremely fast switching/high-frequency applications me use ho sakta hai; isme negative differential resistance region milta hai. [Q9]

3. Superconductivity & High-Temperature Superconductors

- Ideal superconductor critical temperature ke niche zero electrical resistance show karta hai aur Meissner effect me magnetic field ko bulk se expel karta hai; exam language me effective permeability $\approx$ zero li ja sakti hai. [Q10]
- Room-temperature superconductivity economically revolutionary hoti, kyunki transmission losses aur refrigeration/cooling cost dramatically reduce ho sakte hain. Present practical superconductors generally special low-temperature/pressure conditions demand karte hain. [Q11]
- Source Q12 “highest superconducting temperature” ko disputed mark karta hai. Historical cuprate record around 133 K tha, jabki later high-pressure hydrides ne much higher reported transition temperatures show kiye; therefore question ko year/context ke saath padhein. [Q12]
- High-temperature superconductors ka important class cuprate ceramic oxides hai—copper-oxide layers wale ceramic materials ne conventional metallic superconductors se kaafi higher critical temperatures demonstrate kiye. [Q13]

⚠ EXAM TRAP — Q12 — Time-sensitive / Disputed Record

- ▶ Superconducting “highest temperature” record fixed static GK fact nahi hai; material, pressure aur experimental confirmation ke saath record change hota raha hai.
- ▶ Source itself Q12 ko disputed (*) mark karta hai, isliye kisi ek option ko universal current fact na samjhein.

4. Semiconductor Behaviour, Charge Carriers & Transistor Materials

- Intrinsic semiconductor (Si/Ge) at absolute zero me conduction-band free carriers absent hote hain: conductivity $\rightarrow$ zero aur resistance ideally $\rightarrow$ infinite. [Q14, Q19]

- Semiconductors ka negative temperature coefficient hota hai: temperature badhne par more electron-hole pairs generate hote hain, conductivity increases aur resistance decreases. [Q15–Q16]
- Semiconductor current conduction electrons aur holes dono se hoti hai. Thermal excitation, impact ionization aur photo-excitation electron-hole pairs generate kar sakte hain. [Q17–Q18]
- Doping/impurity injection carrier concentration ko control karta hai—n-type me extra electrons, p-type me extra holes—but source Q18 ke context me ye electron-hole “pair generation” technique nahi maana gaya hai. [Q18]
- Germanium aur silicon classic elemental semiconductors hain; transistor fabrication me dono use hue, lekin modern electronics/ICs me silicon most common hai due to abundance, thermal advantages aur useful SiO₂ oxide layer. [Q20–Q25]
- Quartz aur ordinary ceramics given source context me insulators hain; Q23 ka correct semiconductor set Silicon + Germanium hai. [Q23]
- Source Q26 ke according burnt coal/Jhama ash se recoverable germanium valuable semiconductor element hai, jise transistor/electronics applications se link kiya gaya hai. [Q26]



EXAM TRAP — Q23 & Q26 — Source-context Statements

- ▶ “Ceramics are not semiconductors” Q23 ke listed-material context ka answer hai; semiconductor ceramics bhi exist karte hain, so statement ko universal rule na banayein.
- ▶ Jhama export-ban/germanium statement old exam-policy context ka GK fact hai; policy status ko contemporary fact samajhne se pehle question-year dekhein.

✓ Source coverage: Q1–Q26